ExamGOAL
PYQ PDF

Free · With answer key

GATE ECE · ENGINEERING MATHEMATICS · CHAPTER-WISE PYQS

Calculus

GATE ECE Engineering Mathematics · Previous Year Questions with Answer Key

59	29	1994 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

Marks 1	31 questions
Marks 2	28 questions
Answer key	at the end

Practise online

Every question with its full solution, free on **questions.examside.com**

1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.
3. **Understand**
   Scan the QR code beside any question for its full solution.

Marks 1 31 questions

Q1. GATE ECE 2025

Open solution

Consider the following series:

(i) $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$

(ii) $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$

(iii) $\sum_{n=1}^{\infty} \frac{1}{n!}$

(A) Only (ii) converges

(B) Only (ii) and (iii) converge

(C) Only (iii) converges

(D) All three converge

Q2. GATE ECE 2025

Open solution

Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined as

$f(x) = 2x^3 - 3x^2 - 12x + 1$

Which of the following statements is/are correct?

(Here, $\mathbb{R}$ is the set of real numbers.)

(A) f has no global maximizer

(B) f has no global minimizer

(C) $x = -1$ is a local minimizer of f

(D) $x = 2$ is a local maximizer of f

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

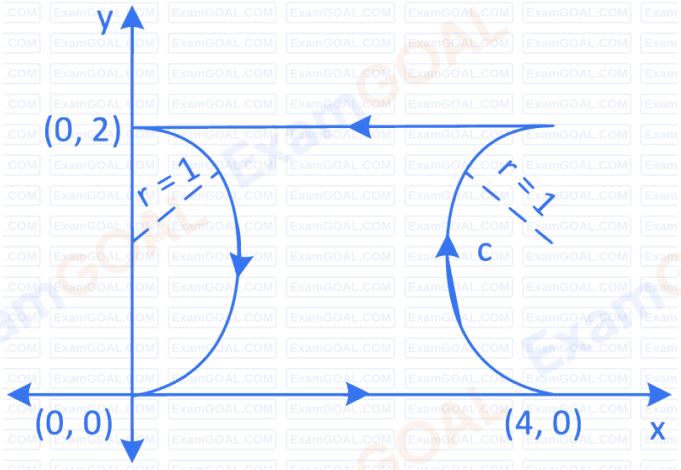
Q3. GATE ECE 2022

Open solution

Consider the two-dimensional vector field $\overrightarrow{F}(x, y) = x\overrightarrow{i} + y\overrightarrow{j}$, where $\overrightarrow{i}$ and $\hat{j}$ denote the unit vectors along the x-axis and the y-axis, respectively. A contour C in the x-y plane, as shown in the figure, is composed of two horizontal lines connected at the two ends by two semicircular arcs of unit radius. The contour is traversed in the counter-clockwise sense. The value of the closed path integral

$$\oint_C \overrightarrow{F}(x, y) \cdot (dx\overrightarrow{i} + dy\overrightarrow{j})$$

is _____.



(A) 0 (B) 1 (C) $8 + 2\pi$ (D) -1

Q4. GATE ECE 2020

Open solution

The partial derivative of the function

$$f(x, y, z) = e^{1-x\cos y} + xze^{\frac{-1}{(1+y^2)}}$$

with respect to x at the point $(1, 0, e)$ is

(A) 1 (B) $\frac{1}{e}$ (C) 0 (D) -1

Q5. GATE ECE 2019

Open solution

The families of curves represented by the solution of the equation

$$\frac{dy}{dx} = -\left(\frac{x}{y}\right)^n$$

for n = −1 and n = 1 respectively, are

(A) Circles and Hyperbolas
(B) Hyperbolas and Circles
(C) Parabolas and Circles
(D) Hyperbolas and Parabolas

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q6. GATE ECE 2018

Let $f(x, y) = \frac{ax^2+by^2}{xy}$, where a and b are constants. If $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y}$ at x = 1 and y = 2, then the relation between a and b is

(A) $a = \frac{b}{4}$ (B) $a = \frac{b}{2}$ (C) $a = 2b$ (D) $a = 4b$

Open solution

Q7. GATE ECE 2018

Taylor series expansion of $f(x) = \int_0^x e^{-\left(\frac{t^2}{2}\right)} dt$ around x = 0 has the form

f(x) = $a_0 + a_1x + a_2x^2 + \ldots$

The coefficient a_2 (correct to two decimal places) is equal to _____.

Open solution

Q8. GATE ECE 2016 Set 2

As x varies from -1 to 3, which of the following describes the behavior of the function $f(x) = x^3 - 3x^2 + 1$?

(A) $f(x)$ increases monotonically

(B) $f(x)$ increases, then decreases and increases again

(C) $f(x)$ decreases, then increases and decreases again

(D) $f(x)$ increases and then decreases

Open solution

Q9. GATE ECE 2016 Set 2

How many distinct values of x satisfy the equation $sin(x) = x/2$, where x is in radians ?

(A) 1 (B) 2 (C) 3 (D) 4 or more

Open solution

Q10. GATE ECE 2016 Set 1

Given the following statements about a function $f : R \to R$, select the right option:
P : If $f(x)$ is continuous at $x = x_0$, then it is also differentiable at $x = x_0$,
Q : If $f(x)$ is continuous at $x = x_0$, then it may not be differentiable at $x = x_0$,
R : If $f(x)$ is differentiable at $x = x_0$, then it is also continuous at $x = x_0$,

(A) P is true, Q is false, R is false (B) P is false, Q is true, R is true

(C) P is false, Q is true, R is false (D) P is true, Q is false, R is true

Open solution

Q11. GATE ECE 2016 Set 3

The integral $\int_0^1 \frac{dx}{\sqrt{(1-x)}}$ is equal _____.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q12. GATE ECE 2015 Set 3

The value of $\sum_{n=0}^{\infty} n\left(\frac{1}{2}\right)^n$ is _____.

Open solution

Q13. GATE ECE 2015 Set 3

The contour on the $x-y$ plane, where the partial derivative of x^2+y^2 with respect to y is equal to the partial derivative of $6y+4x$ with respect to $'x'$, is

(A) $y=2$ (B) $x=2$ (C) $x+y=4$ (D) $x-y=0$

Open solution

Q14. GATE ECE 2015 Set 1

A function $f(x)=1-x^2+x^3$ is defined in the closed interval $[-1,1]$. The value of x, in the open interval $(-1,1)$ for which the mean value theorem is satisfied, is

(A) $-\frac{1}{2}$ (B) $-\frac{1}{3}$ (C) $\frac{1}{3}$ (D) $\frac{1}{2}$

Open solution

Q15. GATE ECE 2014 Set 4

The series $\sum_{n=0}^{\infty} \frac{1}{n!}$ converges to

(A) $2\ ln\ 2$ (B) $\sqrt{2}$ (C) 2 (D) e

Open solution

Q16. GATE ECE 2014 Set 2

For $0 \le t < \infty$, the maximum value of the function $f(t)=e^{-t}-2e^{-2t}$ occurs at

(A) $t=\log_e 4$ (B) $t=\log_e 2$ (C) $t=0$ (D) $t=\log_e 8$

Open solution

Q17. GATE ECE 2014 Set 2

The value of $\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x$ is

(A) $ln2$ (B) 1.0 (C) e (D) ∞

Open solution

Q18. GATE ECE 2014 Set 3

The maximum value of the function $f(x)=\ln(1+x)-x$ (where $x>-1$) occurs at $x=$_____.

Open solution

Q19. GATE ECE 2014 Set 3

If $z=xy\,ln(xy)$, then

(A) $x\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=0$ (B) $y\frac{\partial z}{\partial x}=x\frac{\partial z}{\partial y}$

(C) $x\frac{\partial z}{\partial x}=y\frac{\partial z}{\partial y}$ (D) $y\frac{\partial z}{\partial x}+x\frac{\partial z}{\partial y}=0$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q20. GATE ECE 2010

If $e^y = x^{1/x}$ then y has a

(A) maximum at $x = e$
(B) minimum $x = e$
(C) maximum at $x = e^{-1}$
(D) minimum $x = e^{-1}$

Open solution

Q21. GATE ECE 2008

For real values of x, the minimum value of function $f(x) = e^x + e^{-x}$ is

(A) 2
(B) 1
(C) 0.5
(D) 0

Open solution

Q22. GATE ECE 2008

Which of the following function would have only odd powers of x in its Taylor series expansion about the point $x = 0$?

(A) $\sin(x^3)$
(B) $sin(x^2)$
(C) $cos(x^3)$
(D) $cos(x^2)$

Open solution

Q23. GATE ECE 2007

$\underset{\theta \to 0}{Lim} \frac{\sin(\theta/2)}{\theta}$ is

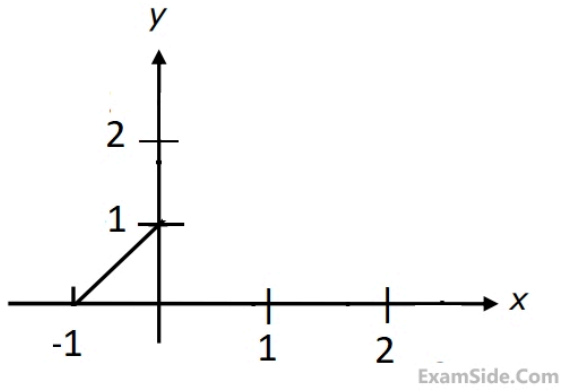
(A) 0.5
(B) 1
(C) 2
(D) not defined

Open solution

Q24. GATE ECE 2007

The following plot shows a function y which varies linearly with x. The value of the integral

$I = \int_1^2 y\, dx$



(A) 1
(B) 2.5
(C) 4
(D) 5

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q25. GATE ECE 2007

For the function e^{-x}, the linear approximation around $x = 2$ is

(A) $(3 - x)e^{-2}$
(B) $1 - x$
(C) $\left[3 + 2\sqrt{2} - \left(1 + \sqrt{2}\right) + x\right]e^{-2}$
(D) e^{-2}

Open solution

Q26. GATE ECE 2007

For $|x| << 1$, $\cot\, h\,(x)$ can be approximated as

(A) x
(B) x^2
(C) $\frac{1}{x}$
(D) $\frac{1}{x^2}$

Open solution

Q27. GATE ECE 2005

The value of the integral $1 = \frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{-x^2/8}\, dx$ is _____.

(A) 1
(B) π
(C) 2
(D) 2π

Open solution

Q28. GATE ECE 1997

The curve given by the equation $x^2 + y^2 = 3axy$ is

(A) Symmetrical about x-axis
(B) Symmetrical about y-axis
(C) Symmetrical about the line $y = x$
(D) Tangential to $x = y = a/3$

Open solution

Q29. GATE ECE 1995

By reversing the order of integration $\int_0^2 \int_{x^2}^{2x} f\,(x, y)dy\,dx$ may be represented as _____.

(A) $\int_0^2 \int_{x^2}^{2x} f\,(x, y)dx\,dy$
(B) $\int_0^2 \int_{y}^{\sqrt{y}} f\,(x, y)dx\,dy$
(C) $\int_0^4 \int_{y/2}^{\sqrt{y}} f\,(x, y)dx\,dy$
(D) $\int_{x^2}^{2x} \int_0^2 f\,(x, y)dx\,dy$

Open solution

Q30. GATE ECE 1995

The third term in the taylor's series expansion of e^x about $'a'$ would be _____.

(A) $e^a\,(x - a)$
(B) $\frac{e^a}{2}(x - a)^2$
(C) $\frac{e^a}{2}$
(D) $\frac{e^a}{6}(x - a)^3$

Open solution

Q31. GATE ECE 1994

The function $y = x^2 + \frac{250}{x}$ at $x = 5$ attains

(A) Maximum
(B) Minimum
(C) Neither
(D) 1

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Marks 2 28 questions

Q32. GATE ECE 2026

Open solution

Consider the two series, S_A and S_B, where

$$S_A = \sum_{n=1}^{\infty} \frac{n^2}{2^n}$$

$$S_A = 1 + \frac{1}{2} + \frac{1}{8} + \frac{1}{16} + \frac{1}{64} + \frac{1}{128} + \frac{1}{512} + \ldots$$

Which of the following statements is correct for the two given series?

(A) Both S_A and S_B converge.

(B) Neither S_A nor S_B converges.

(C) S_A converges but S_B does not converge.

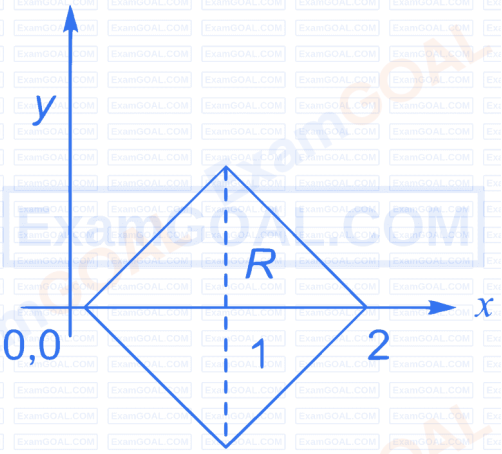
(D) S_B converges but S_A does not converge.

Q33. GATE ECE 2026

Open solution

Consider the square region R in the $X - Y$ plane as shown with the dark shading in the Figure. The value of $\iint_R (x^2 + y^2 - 1)dxdy$ is ____ .

(rounded off to two decimal places)



Q34. GATE ECE 2025

Open solution

Consider a non-negative function $f(x)$ which is continuous and bounded over the interval $[2, 8]$. Let M and m denote, respectively, the maximum and the minimum values of $f(x)$ over the interval.

Among the combinations of α and β given below, choose the one(s) for which the inequality

$\beta \leq \int_2^8 f(x)dx \leq \alpha$

is guaranteed to hold.

(A) $\beta = 5\,\mathrm{m}, \alpha = 7\mathrm{M}$

(B) $\beta = 6\,\mathrm{m}, \alpha = 5\mathrm{M}$

(C) $\beta = 7\,\mathrm{m}, \alpha = 6\mathrm{M}$

(D) $\beta = 7\,\mathrm{m}, \alpha = 5\mathrm{M}$

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q35. GATE ECE 2024

Open solution

Consider the Earth to be a perfect sphere of radius R. Then the surface area of the region, enclosed by the 60°N latitude circle, that contains the north pole in its interior is _____.

(A) $\left(2-\sqrt{3}\right)\pi R^2$

(B) $\frac{\left(\sqrt{2}-1\right)\pi R^2}{2}$

(C) $\frac{2\pi R^2}{3}$

(D) $\frac{\left(2+\sqrt{3}\right)\pi R^2}{8\sqrt{2}}$

Q36. GATE ECE 2023

Open solution

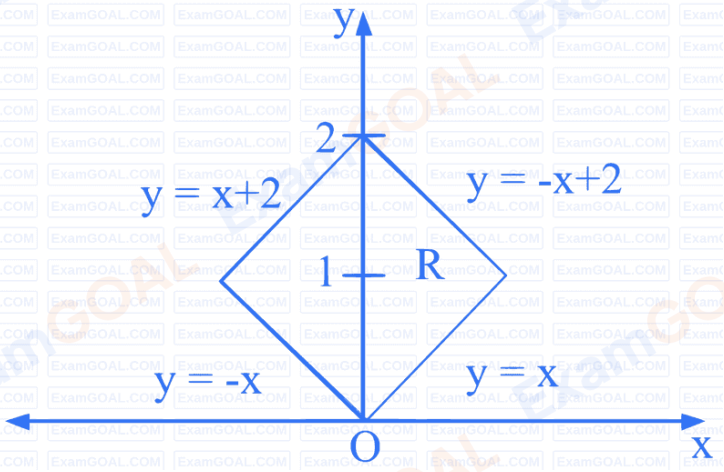
The value of the line integral $\int_P^Q (z^2dx + 3y^2dy + 2xz\,dz)$ along the straight line joining the points $P(1,1,2)$ and $Q(2,3,1)$ is

(A) 20 (B) 24 (C) 29 (D) -5

Q37. GATE ECE 2023

Open solution

The value of the integral $\iint_R xy\,dx\,dy$ over the regioin R, given in the figure, is _____ (rounded off to the nearest integer).



Q38. GATE ECE 2022

Open solution

The function f(x) = 8log$_e$ x $-$ x^2 + 3 attains its minimum over the interval [1, e] at x = _____.

(Here log$_e$ x is the natural logarithm of x.)

(A) 2 (B) 1 (C) e (D) $\frac{1+e}{2}$

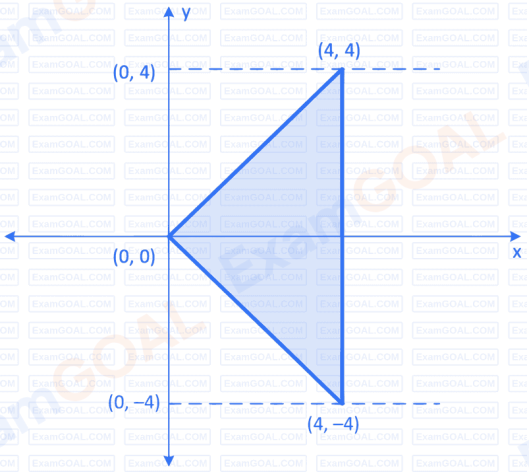
Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q39. GATE ECE 2022

The value of the integral

$$\iint_D 3(x^2 + y^2)dx\,dy,$$

where D is the shaded triangular region shown in the diagram, is _____ (rounded off to the nearest integer).



Open solution

Q40. GATE ECE 2018

Let r = x^2 + y - z and z^3 - xy + yz + y^3 = 1. Assume that x and y are independent variables. At (x, y, z) = (2, -1, 1), the value (correct to two decimal places) of $\frac{\partial r}{\partial x}$ is _____.

Open solution

Q41. GATE ECE 2017 Set 2

The minimum value of the function $f(x) = \frac{1}{3}x\left(x^2 - 3\right)$ in the interval $-100 \leq x \leq 100$ occurs at $x =$ _____.

Open solution

Q42. GATE ECE 2017 Set 2

The values of the integrals $\int_0^1 \left(\int_0^1 \frac{x-y}{(x+y)^3}dy\right)dx$ and $\int_0^1 \left(\int_0^1 \frac{x-y}{(x+y)^3}dx\right)dy$ are

(A) same and equal to 0.5

(B) same and equal to -0.5

(C) 0.5 and -0.5, respectively

(D) -0.5 and -0.5, respectively

Open solution

Q43. GATE ECE 2017 Set 1

Let $f(x) = e^{x+x^2}$ for real x. From among the following. Choose the Taylor series approximation of $f(x)$ around $x = 0$, which includes all powers of x less than or equal to 3.

(A) $1 + x + x^2 + x^3$

(B) $1 + x + \frac{3}{2}x^2 + x^3$

(C) $1 + x + \frac{3}{2}x^2 + \frac{7}{6}x^3$

(D) $1 + x + 3x^2 + 7x^3$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q44. GATE ECE 2017 Set 1

A three dimensional region R of finite volume is described by $x^2 + y^2 \leq z^3,\ 0 \leq z \leq 1$
Where x, y, z are real. The volume of R correct to two decimal places is _____.

Open solution

Q45. GATE ECE 2016 Set 1

The integral $\frac{1}{2\pi} \int\int_D (x + y + 10) dx dy$ where D denotes the disc: $x^2 + y^2 \leq 4$, evaluates to _____.

Open solution

Q46. GATE ECE 2016 Set 1

The region specified by
$\{(\rho, \varphi, \mathrm{Z}) : 3 \leq \rho \leq 5,\ \frac{\pi}{8} \leq \phi \leq \frac{\pi}{4},\ 3 \leq z \leq 4.5\}$ in cylindrical coordinates has volume of _____.

Open solution

Q47. GATE ECE 2016 Set 3

A triangle in the $xy-$plane is bounded by the straight lines $2x = 3y, y = 0$ and $x = 3$. The volume above the triangle and under the plane $x + y + z = 6$ is _____.

Open solution

Q48. GATE ECE 2015 Set 2

The value of the integral $\int_{-\infty}^{\infty} 12\ \cos(2\pi t) \frac{\sin(4\pi t)}{4\pi t} dt$ is _____.

Open solution

Q49. GATE ECE 2015 Set 1

The maximum area (in square units) of a rectangle whose vertices lie on the ellipse $x^2 + 4y^2 = 1$ is

Open solution

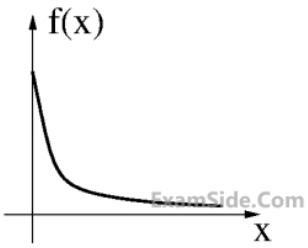
Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q50. GATE ECE 2015 Set 1

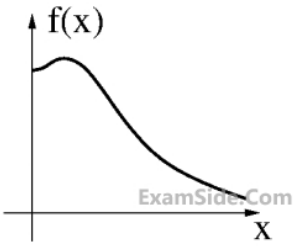
Which one of the following graphs describes the function? $f(x) = e^{-x}\left(x^2 + x + 1\right)$?

Open solution

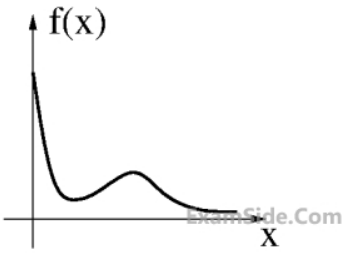
(A)



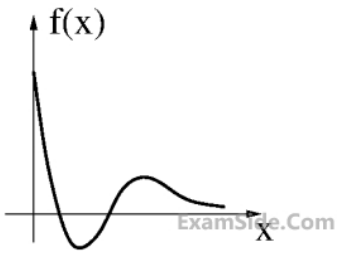
(B)



(C)



(D)



Q51. GATE ECE 2014 Set 1

The Taylor series expansion of $3\ sin\ x + 2cos\ x$ is

Open solution

(A) $2 + 3x - x^2 - \frac{x^3}{2} + - - -$

(B) $2 - 3x + x^2 - \frac{x^3}{2} + - - -$

(C) $2 + 3x + x^2 + \frac{x^3}{2} + - - -$

(D) $2 - 3x - x^2 + \frac{x^3}{2} + - - -$

Q52. GATE ECE 2014 Set 1

The volume under the surface $z(x, y) = x + y$ and above the triangle in the xy plane defined by $\{0 \leq y \leq x$ and $0 \leq x \leq 12\}$ is _____.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q53. GATE ECE 2014 Set 4

For a right angled triangle, if the sum of the lengths of the hypotenuse and a side is kept constant, in order to have maximum area of the triangle, the angle between the hypotenuse and the side is

(A) 12° (B) 36° (C) 60° (D) 45°

Open solution

Q54. GATE ECE 2014 Set 3

The maximum value of $f(x) = 2x^3 - 9x^2 + 12x - 3$ in the interval $0 \leq x \leq 3$ is _____.

Open solution

Q55. GATE ECE 2009

The Taylor series expansion of $\frac{\sin x}{x-\pi}$ at $x = \pi$ is given by

(A) $1 + \frac{(x-\pi)^2}{3!} + - - -$ (B) $-1 - \frac{(x-\pi)^2}{3!} + - - -$

(C) $1 - \frac{(x-\pi)^2}{3!} + - - -$ (D) $-1 + \frac{(x-\pi)^2}{3!} + - - -$

Open solution

Q56. GATE ECE 2008

In the Taylor series expansion of $e^x + \sin x$ about the point $x = \pi$, the coefficient of $(x = \pi)^2$ is

(A) e^π (B) $0.5\ e^\pi$ (C) $e^\pi + 1$ (D) $e^\pi - 1$

Open solution

Q57. GATE ECE 2008

The value of the integral of the function $g(x, y) = 4x^3 + 10y^4$ along the straight line segment from the point $(0, 0)$ to the point $(1, 2)$ in the xy-plane is

(A) 33 (B) 35 (C) 40 (D) 56

Open solution

Q58. GATE ECE 2007

Consider the function $f(x) = x^2 - x - 2$. The maximum value of $f(x)$ in the closed interval $[-4, 4]$

(A) 18 (B) 10 (C) -2.25 (D) indeterminate

Open solution

Q59. GATE ECE 2000

$$\int_0^{\pi/2} \int_0^{\pi/2} \sin(x + y)\, dx\, dy$$

(A) 0

(B) π

(C) $\pi/2$

(D) 2

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Answer Key

Marks 1

1. **B**	2. **A, B**	3. **A**	4. **C**	5. **B**
6. **D**	7. **-0.01 to 0.01**	8. **B**	9. **C**	10. **B**
11. **2**	12. **2**	13. **A**	14. **B**	15. **D**
16. **A**	17. **C**	18. **0**	19. **C**	20. **A**
21. **A**	22. **A**	23. **A**	24. **B**	25. **A**
26. **C**	27. **A**	28. **C**	29. **C**	30. **B**
31. **B**				

Marks 2

32. **A**	33. **(0.66) (0.60 to 0.70)**	34. **A**	35. **A**	36. **B**
37. **0**	38. **B**	39. **512**	40. **4.4 to 4.6**	41. **-100**
42. **C**	43. **C**	44. **0.79**	45. **20**	46. **4.71**
47. **10**	48. **3**	49. **1**	50. **B**	51. **A**
52. **864**	53. **C**	54. **6**	55. **D**	56. **A**
57. **A**	58. **A**	59. **D**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

ExamGOAL
PYQ PDF

Your Exam. Our Responsibility.

From first chapter to exam day.

Practise with the questions your exam has already asked, then test yourself under real exam conditions.

PAST YEAR QUESTIONS

Every past paper, chapter-wise.

Previous year questions sorted the way you study, each with a full solution.

- Chapter-wise and year-wise
- Weightage and topic tags on every chapter
- Your progress on each one, at a glance

MOCK TESTS

Mock tests with a real exam feel.

Practise under the same pressure as the real paper, on your phone or on the big screen.

- Timed sections, just like exam day
- Mark questions for review and come back

ANALYSIS

Know your score. Know your rank.

A full report after every test, so you know exactly where you stand.

- Your percentile among everyone who took it
- Predicted real-exam percentile and All India Rank
- Subject breakdown and time analysis

PRACTICE

Solve. Check. Understand.

Learn from every question, not just the ones you get right.

- Instant answers as you practise
- Step-by-step explanations
- See how everyone else answered

NCERT Exemplar
With full solutions

Exam syllabus
Subject and chapter-wise

Bookmarks
Organised in groups

Your notes
On any question

ExamGOAL
LEARNING APP

ExamGOAL: Exam Preparation App

One app for every exam. Free trial for new users, on Android and the web.

GET IT ON Google Play

Scan to get the app

www.examgoal.com
Practice and mock tests

questions.examside.com
Free solutions

pdf.questions.examgoal.com
Free PYQ PDFs